

CSES — Additional Problems II Reference

All tasks: 1 s, 512 MiB. Modulo $10^9 + 7$ only where explicitly stated.

1148 — Maximum Building II

Given an $n \times m$ forest grid of empty cells `.` and trees `*`, count placements of an all-empty axis-aligned rectangle for *every exact size* $h \times w$.

In: `n` `m`; then n grid rows

Out: n rows; row h contains the counts for widths $w = 1, \dots, m$

Constraints: $1 \leq n, m \leq 1000$.

1157 — Mex Grid Queries

An infinite positive-index grid is filled so cell (y, x) is the smallest nonnegative integer absent both to its left in row y and above it in column x . Find the value at one queried cell. **In:** `y` `x`

Out: value at (y, x)

Constraints: $1 \leq y, x \leq 10^9$.

1159 — Book Shop II

Book i costs h_i , gives s_i pages, and has at most k_i copies available. With total cost at most x , maximize purchased pages. **In:** `n` `x`; prices h ; pages s ; copy limits k

Out: maximum total pages

Constraints: $n \leq 100$, $x \leq 10^5$, $h_i, s_i, k_i \leq 1000$.

1161 — Stick Divisions

Start with one stick of length x . A move splits one current stick into two and costs the original stick length; end with the multiset of target lengths $d_1, \dots, d_n$. Minimize total cost. **In:** `x` `n`; $d_1 \dots d_n$

Out: minimum total split cost

Constraints: $x \leq 10^9$, $n \leq 2 \cdot 10^5$, $\sum d_i = x$. Use 64-bit.

1189 — Food Division

Children form a cycle. Child i initially has a_i food and must end with b_i ; one step transfers one unit to either neighbor. The total food is already correct. Minimize transfers. **In:** `n`; $a_1 \dots a_n$; $b_1 \dots b_n$

Out: minimum number of unit transfers

Constraints: $n \leq 2 \cdot 10^5$, $a_i, b_i \leq 10^6$. Use 64-bit.

1665 — Coding Company

Partition the n coders into teams. A team's penalty is $\max t - \min t$; count divisions whose sum of team penalties is at most x . **In:** `n` `x`; skills $t_1 \dots t_n$

Out: number of valid divisions mod $10^9 + 7$

Constraints: $n \leq 100$, $x \leq 5000$, $0 \leq t_i \leq 100$.

1698 — Swap Round Sorting

Sort a permutation in the minimum number of *swap rounds*; within one round, choose any number of distinct pairs and swap each chosen pair. Also construct the swaps in every round. **In:** `n`; permutation p

Out: minimum round count; for each round, its swaps as index pairs

Constraints: $n \leq 2 \cdot 10^5$.

1706 — School Excursion

Each given pair of children insists on visiting the same attraction, so every connected component must stay together. For each $i = 1, \dots, n$, decide whether exactly i children can be assigned to Korkeasaari. **In:** `n` `m`; then m same-attraction pairs

Out: length- n bit string; bit i is 1 iff size i is possible

Constraints: $n, m \leq 10^5$.

1709 — Coin Grid

An $n \times n$ binary coin grid can be cleared by a move that removes all remaining coins in one chosen row or column. Find the minimum number of moves and output any optimal set of rows/columns. **In:** `n`; then n rows over `.`, `o`

Out: minimum k , then k moves 1 `r` (row) or 2 `c` (column)

Constraints: $1 \leq n \leq 100$.

2075 — Reversal Sorting

Sort a permutation by reversing contiguous subarrays. Any construction using at most n reversals is accepted; the number of reversals need not be minimum. **In:** `n`; permutation p

Out: $k \leq n$, then reversal intervals $[a, b]$

Constraints: $n \leq 2 \cdot 10^5$.

2115 — Bit Substrings

For each $k = 0, 1, \dots, n$, count the *nonempty* substrings of a bit string that contain exactly k ones. **In:** binary string s

Out: $n + 1$ counts for $k = 0, \dots, n$

Constraints: $|s| \leq 2 \cdot 10^5$. Counts need 64-bit.

2131 — Grid Puzzle II

Choose grid cells so row i contains exactly a_i chosen cells and column j exactly b_j . Cell (i, j) yields c_{ij} coins; maximize the collected sum and construct the chosen cells, or detect infeasibility. **In:** `n`; row quotas a ; column quotas b ; then coin matrix c

Out: maximum sum and an `X/.` grid, or `-1`

Constraints: $n \leq 50$, $a_i, b_j \leq n$, $c_{ij} \leq 1000$; $\sum a_i = \sum b_j$.

2132 — Increasing Array II

Change each array value by unit increments/decrements until the sequence is nondecreasing. Minimize the total absolute change. **In:** `n`; $x_1 \dots x_n$

Out: minimum number of unit changes

Constraints: $n \leq 2 \cdot 10^5$, $x_i \leq 10^9$. Use 64-bit.

2174 — Removing Digits II

Starting from n , repeatedly subtract any one decimal digit currently appearing in the number. Find the minimum steps to reach 0. **In:** `n`

Out: minimum number of steps

Constraints: $1 \leq n \leq 10^{18}$. Use 64-bit.

2180 — Coin Arrangement

A $2 \times n$ grid contains exactly $2n$ coins total, possibly multiple per cell. Move individual coins one orthogonal step per operation until every cell has exactly one; minimize total moves. **In:** `n`; then the two rows of coin counts

Out: minimum number of moves

Constraints: $1 \leq n \leq 10^5$. Use 64-bit.

2402 — Two Stacks Sorting

Process a permutation from left to right using two stacks: move the first remaining input value onto one stack, or pop a stack value to the output. The final output must be $1, 2, \dots, n$. **In:** $\mathbf{n}$; input permutation
Out: for each input number, a chosen stack 1 or 2; or IMPOSSIBLE
Constraints: $n \leq 2 \cdot 10^5$.

2426 — Programmers and Artists

Each applicant has programming skill x and artistic skill y . Hire exactly a programmers and b artists, with no applicant used twice, maximizing total skill. **In:** $\mathbf{a} \ \mathbf{b} \ \mathbf{n}$; then n pairs $\mathbf{x} \ \mathbf{y}$
Out: maximum total skill
Constraints: $n \leq 2 \cdot 10^5$, $a + b \leq n$, $x, y \leq 10^9$. Use 64-bit.

2430 — Binary Subsequences

Construct a shortest bit string having exactly n distinct *nonempty* subsequences (identical resulting strings count once). Any minimum-length construction is accepted. **In:** $\mathbf{n}$
Out: any minimum-length valid bit string
Constraints: $1 \leq n \leq 10^6$.

2432 — Grid Puzzle I

Choose cells of an $n \times n$ grid so row i has exactly a_i chosen cells and column j exactly b_j . Construct any feasible $\mathbf{X}/.$ grid. **In:** $\mathbf{n}$; row quotas a ; column quotas b
Out: a valid grid, or $\mathbf{-1}$ if impossible
Constraints: $n \leq 50$, $0 \leq a_i, b_j \leq n$.

3108 — K Subset Sums I

Consider all 2^n subsets, including the empty subset. Output the k smallest subset sums in nondecreasing order; equal sums from different subsets appear with multiplicity. **In:** $\mathbf{n} \ \mathbf{k}$; $x_1 \dots x_n$
Out: k smallest subset sums
Constraints: $n \leq 2 \cdot 10^5$, $k \leq 2 \cdot 10^5$, $|x_i| \leq 10^9$. Use 64-bit sums.

3109 — K Subset Sums II

Consider all subsets containing exactly m array elements. Output the k smallest sums in nondecreasing order, retaining multiplicity between different subsets. **In:** $\mathbf{n} \ \mathbf{m} \ \mathbf{k}$; $x_1 \dots x_n$
Out: k smallest size- m subset sums
Constraints: $m < n \leq 2 \cdot 10^5$, $k \leq 2 \cdot 10^5$, $|x_i| \leq 10^9$. Use 64-bit sums.

3159 — Replace with Difference

Repeatedly choose two *current numbers* a, b , delete both, and insert $|a - b|$. Construct $n - 1$ operations so the sole remaining value is 0, or report impossibility. **In:** $\mathbf{n}$; $x_1 \dots x_n$
Out: chosen value pairs a, b for every operation, or $\mathbf{-1}$
Constraints: $2 \leq n \leq 1000$, $1 \leq x_i \leq 1000$.

3161 — GCD Subsets

For every $k = 1, \dots, n$, count nonempty subsets of array positions whose selected values have greatest common divisor exactly k . **In:** $\mathbf{n}$; $x_1 \dots x_n$
Out: n answers for $\gcd 1, \dots, n$, modulo $10^9 + 7$
Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x_i \leq n$.

3215 — Bouncing Ball Steps

A ball starts at (1, 1) in an $n \times m$ grid moving diagonally down-right, one cell per step, reflecting whenever it hits a border. For a huge k , find its position and total number of direction changes after k steps. **In:** $\mathbf{t}$; then tests $\mathbf{n} \ \mathbf{m} \ \mathbf{k}$
Out: for each test: row, column, direction-change count
Constraints: $t \leq 1000$, $n, m \leq 10^9$, $k \leq 10^{18}$. Use 64-bit.

3216 — Bouncing Ball Cycle

For the same reflecting diagonal motion, determine when the ball first reaches (1, 1) again and how many distinct grid cells were visited during the cycle. **In:** $\mathbf{t}$; then tests $\mathbf{n} \ \mathbf{m}$
Out: for each test: return time and number of distinct visited cells
Constraints: $t \leq 1000$, $2 \leq n, m \leq 10^9$. Use 64-bit.

3218 — Knight Moves Queries

On the infinite chessboard with positive row/column indices, answer the minimum knight moves from each queried (x, y) to the top-left corner (1, 1). **In:** $\mathbf{q}$; then q positions $\mathbf{x} \ \mathbf{y}$
Out: minimum knight distance for each query
Constraints: $q \leq 10^5$, $x, y \leq 10^9$.

3312 — Grid Coloring II

Every cell initially contains A, B, or C. Change *every* cell to a different letter so that no orthogonally adjacent final cells are equal. Construct any solution. **In:** $\mathbf{n} \ \mathbf{m}$; then n grid rows
Out: a valid recolored grid, or IMPOSSIBLE
Constraints: $n, m \leq 500$.

3401 — Stick Difference

Make exactly k cuts among integer-length sticks, producing $n + k$ positive-integer pieces. For every $k = 1, \dots, m$, minimize $\max(\text{piece}) - \min(\text{piece})$. **In:** $\mathbf{n} \ \mathbf{m}$; lengths $a_1 \dots a_n$
Out: m minima for exactly $k = 1, \dots, m$ cuts
Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $a_i \leq 10^9$; all m cuts are feasible.

3402 — Minimum Cost Pairs

For each $k = 1, \dots, \lfloor n/2 \rfloor$, choose k disjoint pairs of array elements. Pair (a, b) costs $|a - b|$; minimize the total cost. **In:** $\mathbf{n}$; $x_1 \dots x_n$
Out: $\lfloor n/2 \rfloor$ minimum costs
Constraints: $n \leq 2 \cdot 10^5$, $x_i \leq 10^9$. Use 64-bit.

3425 — Same Sum Subsets

Choose two disjoint nonempty subsets of the given positive elements whose sums are equal. Output any valid pair of subsets. **In:** $\mathbf{n}$; $x_1 \dots x_n$
Out: for each subset, its size and elements; or IMPOSSIBLE
Constraints: $3 \leq n \leq 40$, $\sum x_i \leq 2^n - 2$.